7. The graph of $y = e^x$ intersect the y-axis at

(a) (0, 0) (b) (1, 0)
(c) (0, 1) (d) (1, 1)

8. The minimum value of the function $f(x)=|x|$ is

(a) 0 (b) −1
(c) +1 (d) $-\infty$

9. Which one of the following functions has the property $f(x)=f\left(\frac{1}{x}\right)$, provided $x \neq 0$

(a) $f(x)=\frac{x^2-1}{x}$

(b) $f(x)=\frac{1-x^2}{x}$

(c) $f(x) = x$

(d) $f(x)=\frac{x^2+1}{x}$

10. If $f(x)=2^x$ and $g(x) = \frac{1}{2^x}$, then $(fg)(x)$ is

(a) 1 (b) 0
(c) 4^x (d) $\frac{1}{4^x}$

11. Which of the following function is neither even nor odd?

(a) $f(x)=x^3+5$ (b) $f(x)=x^5$
(c) $f(x)=x^{10}$ (d) $f(x)=x^2$

12. $f(x)=-5$, for all $x \in R$ is a

(a) an identity function

(b) modulus function

(c) exponential function

(d) constant function

13. The range of $f(x)=|x|$, for all $x \in R$ is

(a) $(0, \infty)$ (b) $[0, \infty)$
(c) $(-\infty, \infty)$ (d) $[1, \infty)$

14. The graph of $f(x) = e^x$ is identical to that of

(a) $f(x)=a^x, a>1$

(b) $f(x)=a^x, a<1$

(c) $f(x)=a^x, 0<a<1$

(d) $y = ax + b, a \neq 0$

15. If $f(x)=x^2$ and $g(x) = 2x+1$, then $(fg)(0)$ is

(a) 0 (b) 2
(c) 1 (d) 4

16. $\lim\limits_{\theta \to 0} \frac{\tan\theta}{\theta}$ =

(a) 1 (b) ∞
(c) $-\infty$ (d) θ

17. $\lim\limits_{x \to 0} \frac{e^x-1}{x}$ =

(a) e (b) nx^{n-1}
(c) 1 (d) 0

18. For what value of x, $f(x)=\frac{x+2}{x-1}$ is not continuous?

(a) −2 (b) 1
(c) 2 (d) −1

19. If the function $f(x)$ is continuous at $x = a$, then $\lim\limits_{x \to a} f(x)$ is equal to

(a) $f(-a)$ (b) $f\left(\frac{1}{a}\right)$
(c) $2f(a)$ (d) $f(a)$

20. $\frac{d}{dx}\left(\frac{1}{x}\right)$ is equal to

(a) $-\frac{1}{x^2}$ (b) $-\frac{1}{x}$
(c) $\log x$ (d) $\frac{1}{x^2}$

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